

Repeated Exam MA-224, part 2

Course code: MA-224

Course name: Discrete mathematics 2

Date: 24. February 2023

Time: 09:00 – 15:00 (6 hours)

Number of pages: 5

Number of tasks: 12

Permitted equipment: All written material, no communication

Remarks: Present your answers so that it is clear what methods you have used, and how you have used them.
All problems are scored equally.

Problem 1

We consider the following theorem.

$$\text{Th1: } [(p \rightarrow r) \wedge (q \rightarrow r)] \iff [(p \vee q) \rightarrow r]$$

(a) Prove Th1 using truth tables.

(b) Prove Th1 using logical inferences.

For your convenience, here are some possible inference rules.

equivalence between $\rightarrow$ and $\wedge$: $A \rightarrow B \iff \neg A \vee B$

deMorgan's Law for $\wedge$: $\neg(A \wedge B) \iff \neg A \vee \neg B$

deMorgan's Law for $\vee$: $\neg(A \vee B) \iff \neg A \wedge \neg B$

commutativity of $\wedge$: $A \wedge B \iff B \wedge A$

associativity of $\wedge$: $(A \wedge B) \wedge C \iff A \wedge (B \wedge C)$

distributivity of $\wedge$: $(A \vee B) \wedge C \iff (A \wedge C) \vee (B \wedge C)$

commutativity of $\vee$: $A \vee B \iff B \vee A$

associativity of $\vee$: $(A \vee B) \vee C \iff A \vee (B \vee C)$

distributivity of $\vee$: $(A \wedge B) \vee C \iff (A \vee C) \wedge (B \vee C)$

Problem 2

Define the following sequence recursively: $s_i = 3 \cdot i + 10$.

Problem 3

Consider the following EBNF grammar with the start symbol s . Terminals are enclosed in single quotes ('go'). Alternatives are denoted by '|'.

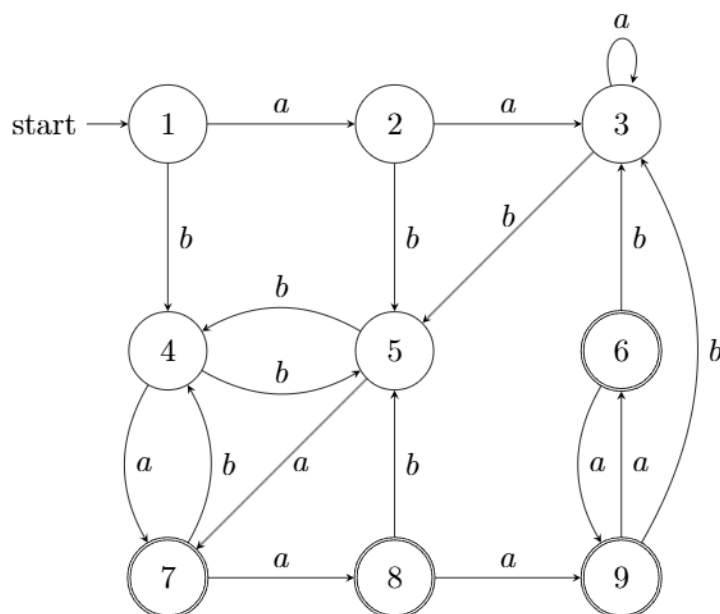
$$s \rightarrow \text{'go'} \mid \text{'stop'} \mid \text{'while'} s \text{'do'} s \mid s \text{' ; ' } s$$

(a) Find a derivation in the EBNF grammar above for the string
while stop do go ; stop.

(b) Is the grammar ambiguous? Give reasons.

Problem 4

Minimize the following DFA.



Problem 5

Is the code below correct related to the following specification?

Specification: $\text{SQRT}(x, \text{res}:\text{NUM}) \text{ res} := \text{res}' \mid \text{res}' * \text{res}' = x$

Code: `float sqrt(x:float) { return (-1) * Math.sqrt(x); }`

Problem 6

We define a relation $=_{nin2}$ between Norwegians such that two Norwegians are related with $=_{nin2}$ if they have the same first two digits of their national identity number. Otherwise, they are not related with $=_{nin2}$.

Prove that $=_{nin2}$ is an equivalence relation, i.e. that $=_{nin2}$ is reflexive, symmetric, and transitive.

You can assume that there is a function $nin2$, which assigns the first two digits of the national identity number (i.e., a number) to each Norwegian N .

Problem 7

A sequence of numbers $a_0, a_1, a_2, \dots$ is said to be periodic with period p if there exists a positive integer $p \geq 1$ such that $a_{n+p} = a_n$ for all $n \geq 0$. In other words, the sequence repeats after p terms.

Find an integer b such that the solution to the recurrence relation

$$\begin{cases} a_n = ba_{n-1} - a_{n-2} & ; n \geq 2 \\ a_0 = 0 \\ a_1 = 1 \end{cases}$$

is periodic.

Problem 8

A non-trivial subgroup $H \subset G$ is a subgroup which is neither G nor $\{e\}$.

Let Σ_5 be the group of permutations of 5 symbols.

- (a) Find a non-trivial subgroup $H_1 \subset \Sigma_5$ which is not abelian.
- (b) Find a non-trivial subgroup $H_2 \subset \Sigma_5$ which is cyclic.
- (c) Find a non-trivial subgroup $H_3 \subset \Sigma_5$ which is abelian.

Problem 9

Let $\phi(n) = |P_n|$ be the Euler totient function.

- (a) Is $\phi(n)$ an increasing function of n ? Explain your answer.
- (b) Prove using the definition of $\phi(n)$ that $\phi(2^k) = 2^{k-1}$, for every positive integer $k \geq 1$.

Problem 10

Let G be a group of order 35 and let $H \subset G$ be a subgroup such that $\{1\} \neq H \neq G$.

Use Lagrange's theorem to prove that the index $[G : H]$ is a prime number.

Problem 11

Give an example of a coding function

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 6}$$

which can be used to correct all transmission errors of weight 1. Explain why your coding function has the desired property.

Problem 12

Given the (3×6) -generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Assume that a 3-bit message is encoded using G and transmitted over a noisy channel, and that the received message is $r = \mathbf{010011}$. Assume that at most one bit error occurs during the transmission of the message.

Compute the syndrome of r and use it to detect and correct any error in the transmission.